

EXPERIMENT - 4

OLAP Operations (Slice, Dice, Drill-up & Drill-down) using Microsoft Power BI

Report



Subject Name: Business Intelligence & Tools
Subject Code: 24CSH-341

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Aim

To perform Online Analytical Processing (OLAP) operations using Microsoft Power BI by creating multidimensional reports and applying Slice, Dice, Drill-up, and Drill-down operations on the Global Superstore dataset for analytical decision-making.

Tools / Software Required

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX

Dataset

Dataset Used: Global Superstore Dataset

The Global Superstore dataset contains business information related to sales, products, customers, regions, categories, segments, and other business dimensions. It is used to perform multidimensional analysis using OLAP operations.

Theory

OLAP

Online Analytical Processing (OLAP) is a Business Intelligence technique used for analyzing data from multiple dimensions. It helps users examine business information from different perspectives and levels of detail.

Power BI supports OLAP-style analysis using:

- Hierarchies
- Filters
- Slicers
- Visual interactions
- Drill-down and drill-up functionality

OLAP Operations

1. Slice:

Slice selects a single value from one dimension to analyze a specific part of the dataset.

2. Dice:

Dice selects multiple dimensions and values simultaneously to obtain a smaller and more specific subset of data.

3. Drill-down:

Drill-down moves from summarized information to more detailed information.

4. Drill-up:

Drill-up moves from detailed information to a higher-level summary.

Procedure

Part A – Load Dataset and Build Report

Step 1: Open Power BI Desktop

Microsoft Power BI Desktop was opened to create the OLAP report.

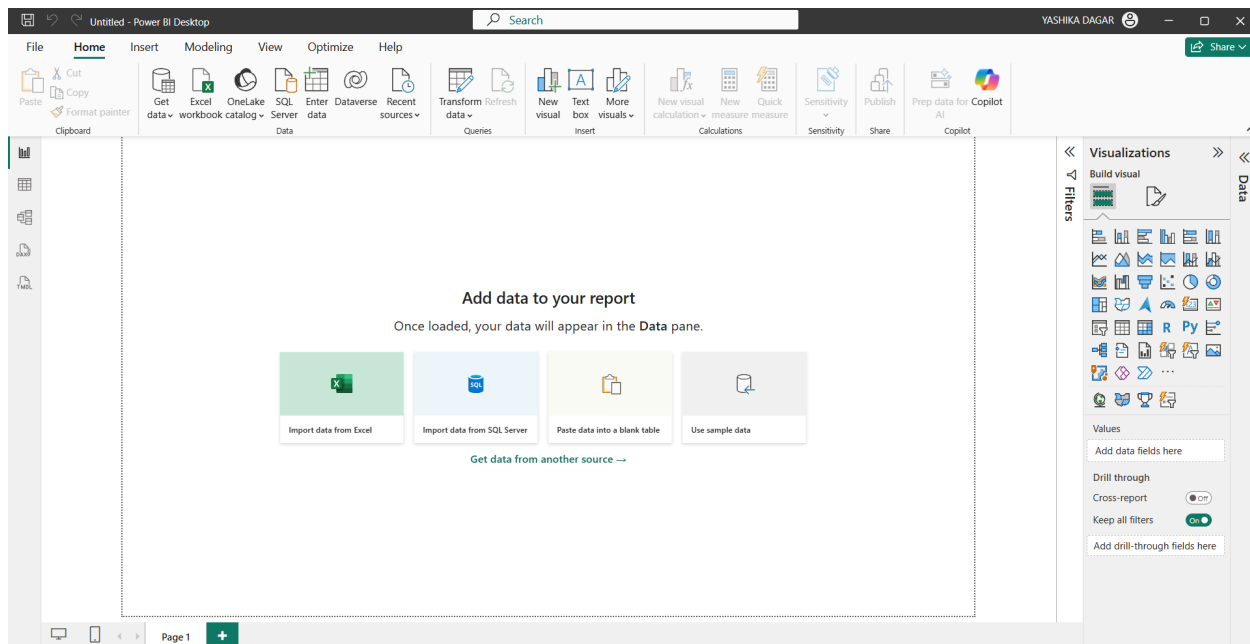


Figure 1: Microsoft Power BI Desktop

Step 2: Load the Dataset

The Global Superstore dataset was imported into Power BI.

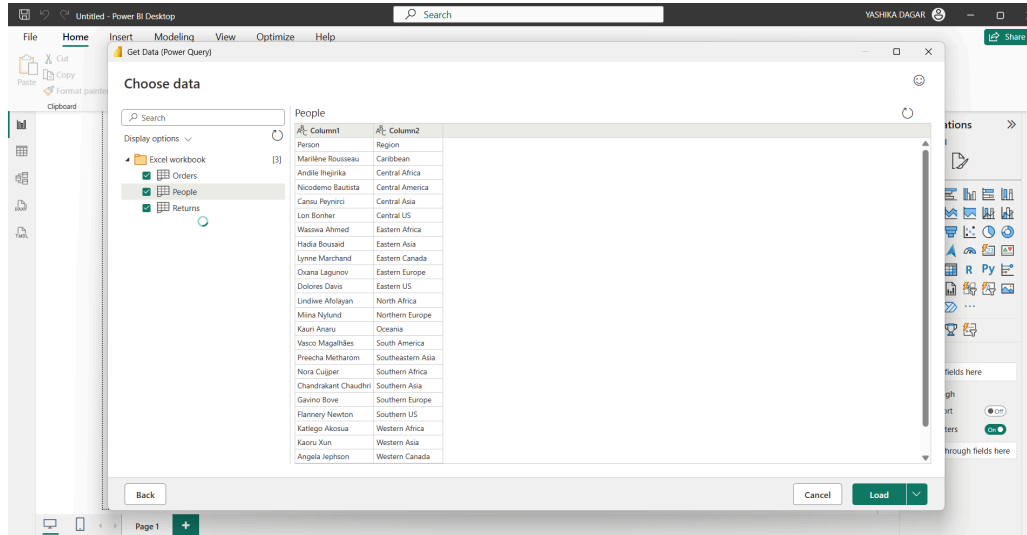


Figure 2: Global Superstore dataset loaded in Power BI

Step 3: Verify Relationships

The relationships between the Orders, Returns, and People tables were checked in the Model view.

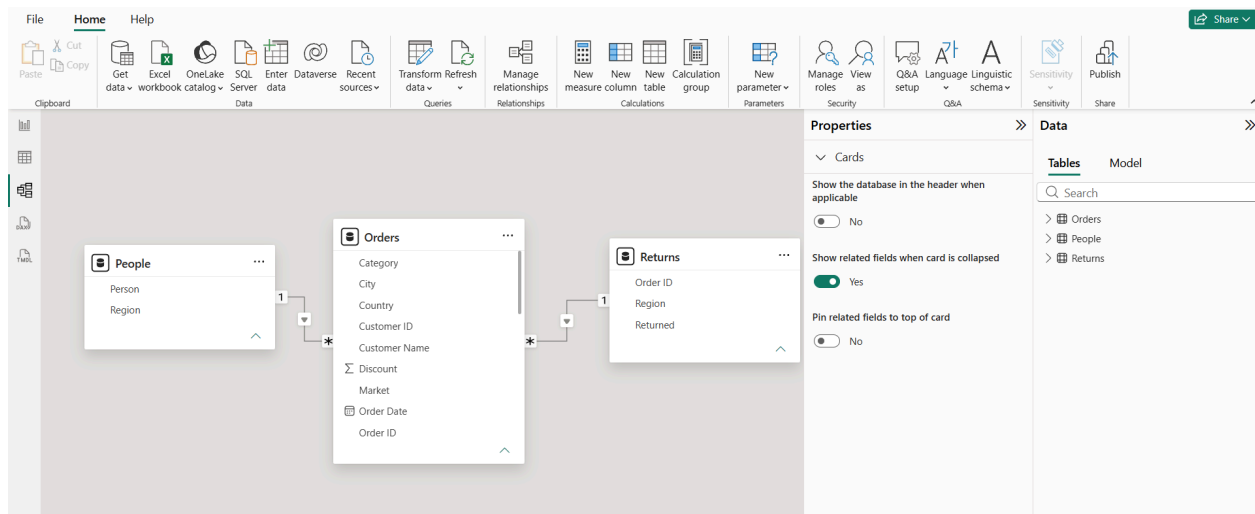


Figure 3: Relationships between Orders, Returns, and People tables

Part A – Report Visuals

The following visuals were created in Power BI:

1. Sales by Region

A visual was created to display sales across different regions.

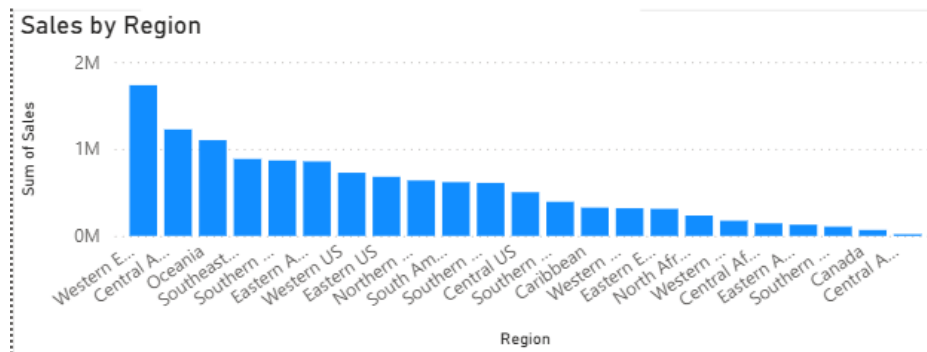


Figure 4: Sales by Region

2. Profit by Category

A visual was created to analyze profit across different product categories.

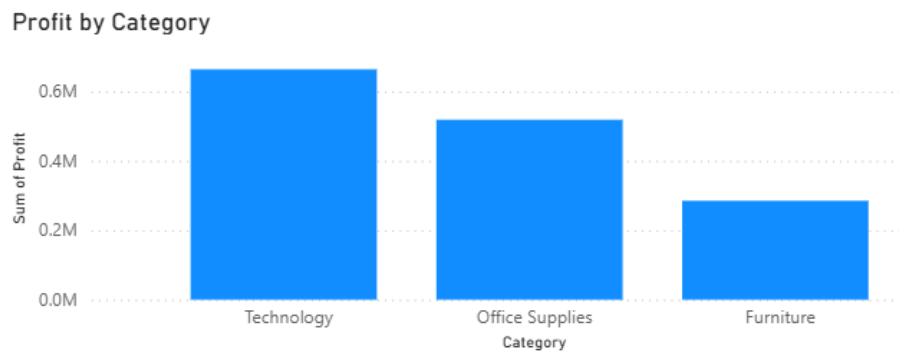


Figure 5: Profit by Category

3. Sales by Segment

A visual was created to compare sales across customer segments.

Sales by Segment



Figure 6: Sales by Segment

4. Sales Trend by Year

A visual was created to observe the change in sales across years.

Sales trend by Year



Figure 7: Sales Trend by Year

Part B – Slice Operation

Step 5: Add Region Slicer

A slicer was added to the report using the **Region** field.

Step 6: Select a Single Region

A single region, such as **West**, was selected from the Region slicer.

Step 7: Observe the Changes

After selecting the region, the report visuals were automatically filtered. Changes in sales, profit, and orders were observed.

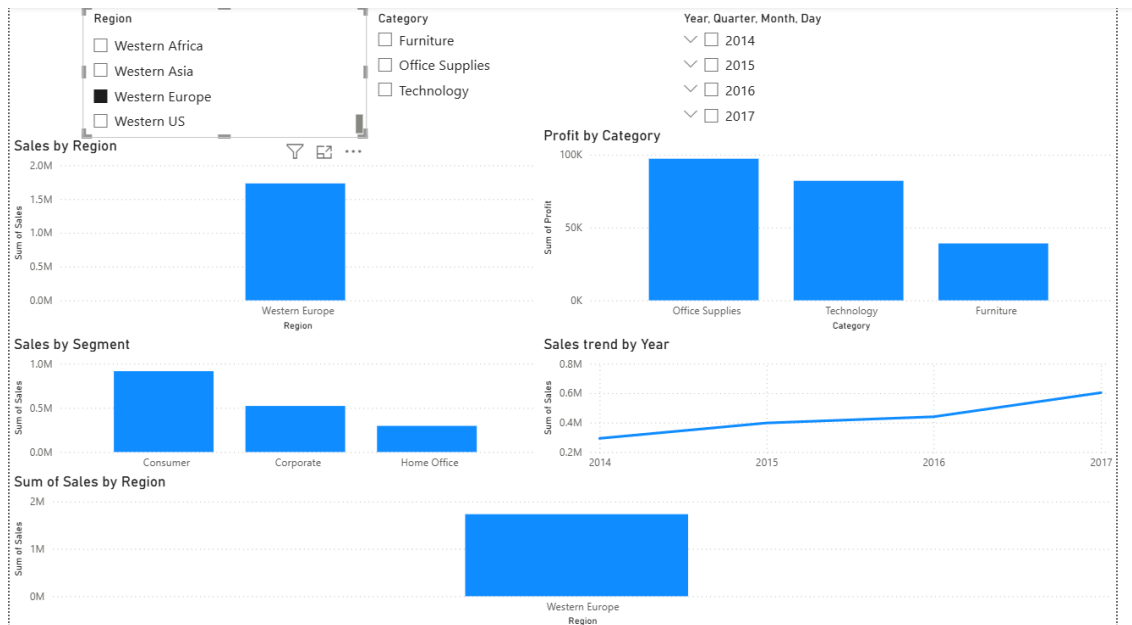


Figure 8: Slice operation using the Region slicer

Observation

The Slice operation reduced the dataset to the selected region. The report visuals displayed information only for the selected region.

Part C – Dice Operation

Step 9: Apply Multiple Filters

Multiple dimensions were selected simultaneously:

- **Region = West**
- **Category = Technology**
- **Year = 2022**

Step 10: Analyze Filtered Metrics

The report was filtered using all three conditions. The resulting sales, profit, and order-related metrics were analyzed.

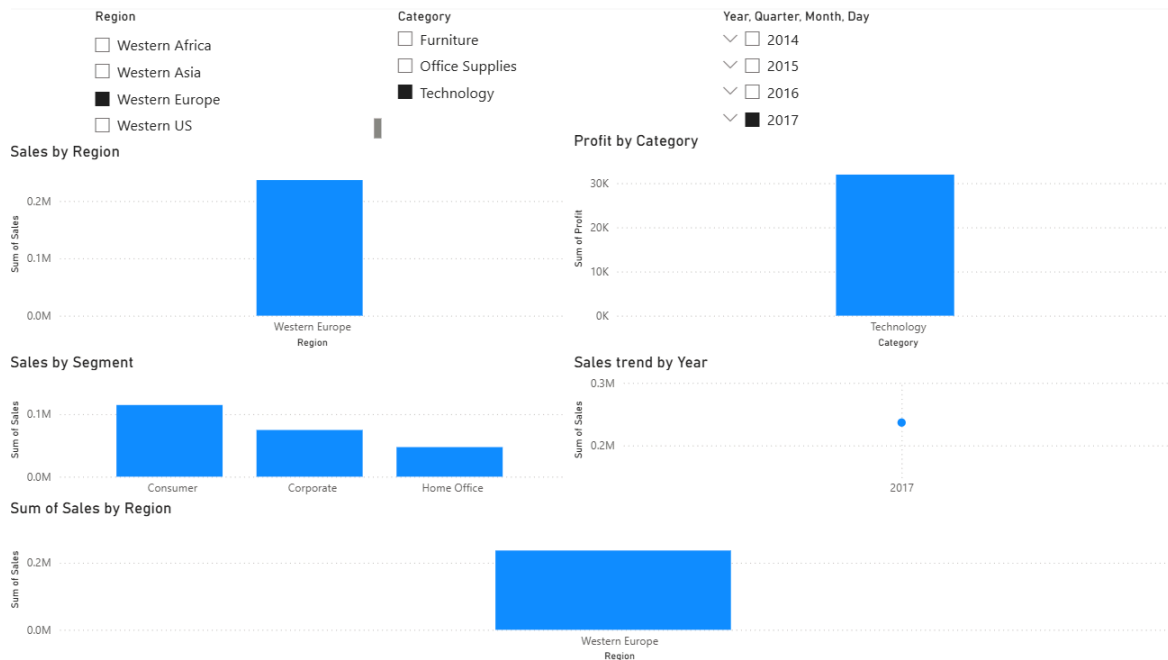


Figure 9: Dice operation using multiple dimensions

Observation

The Dice operation produced a more specific subset of the dataset by applying multiple filters simultaneously.

Part D – Drill-down Operation

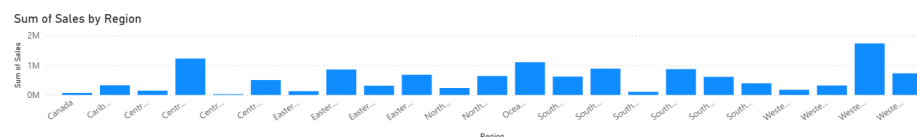
Step 12: Create Hierarchy

A geographical hierarchy was created as:

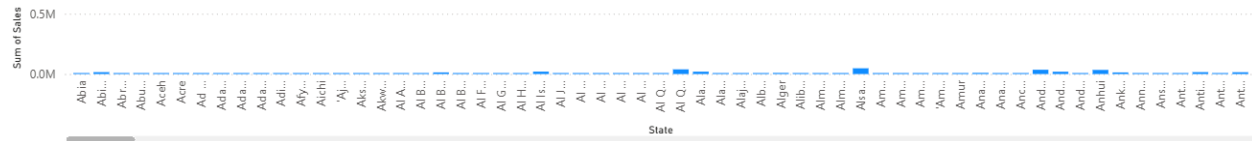
Region → State → City

Step 13: Create a Bar Chart

A bar chart was created using the Region–State–City hierarchy.



Sum of Sales by State



Sum of Sales by City

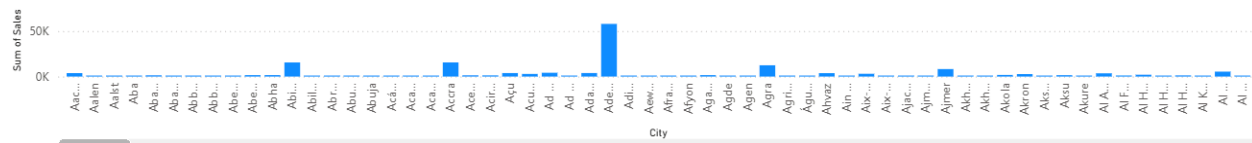


Figure 10: Region–State–City hierarchy

Observation

Drill-down provided more detailed information by moving from a higher-level geographical summary to individual states and cities.

Part E – Drill-up Operation

Step 17: Navigate from City to Region

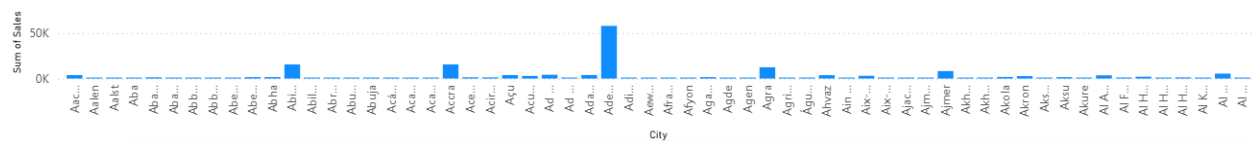
The hierarchy was used in the reverse direction:

City → State → Region

Step 18: Compare Aggregated Values

The aggregated values were compared at the City, State, and Region levels.

Sum of Sales by City



Sum of Sales by State

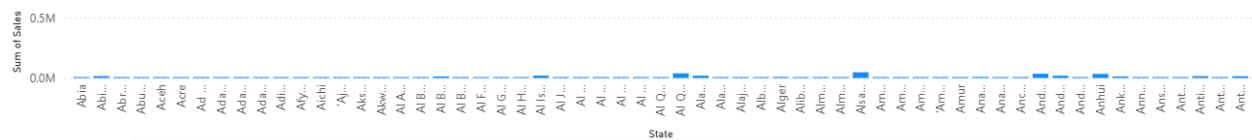


Figure 13: Drill-up from City to State

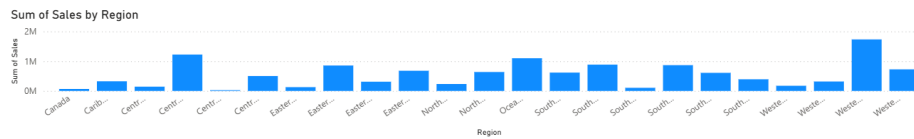


Figure 14: Drill-up from State to Region

Observation

Drill-up summarized detailed information into higher-level categories, making it easier to view overall business performance.

Result

The OLAP operations **Slice, Dice, Drill-down, and Drill-up** were successfully performed using Microsoft Power BI on the Global Superstore dataset.

The report enabled multidimensional analysis of business data from different perspectives and levels of detail.

Learning Outcomes

After completing this experiment:

1. I understood the concepts of OLAP and multidimensional data analysis.
2. I performed Slice and Dice operations using filters and slicers in Power BI.
3. I created a geographical hierarchy and performed Drill-down and Drill-up operations.
4. I interpreted analytical insights generated through OLAP operations.
5. I designed an interactive analytical report for business analysis.

Conclusion

This experiment demonstrated how Microsoft Power BI can be used for OLAP-style multidimensional analysis. Slice and Dice operations helped filter specific portions of the dataset, while Drill-down and Drill-up provided detailed and summarized views of the data. These operations made it easier to explore business information and understand data from multiple perspectives.